

metadata

platform: LinkedIn audience: ML engineers, RAG/LLM evaluation, data engineering
tone: technical but approachable cta: Repository + methodology in the comments / link below.

Your LLM-judge RAG metrics may be replaceable — and one of them may be lying to you.

I ran the four classic RAGAS metrics (LLM-as-a-judge, fully local) on a real agentic-RAG memory system, then compared them against cheap deterministic shortcuts. Spearman agreement over 99 real query/response pairs:

- **AnswerCorrectness** vs. ROUGE-L + embedding cosine → $\rho = \mathbf{0.89}$ (STRONG)
- **AnswerRelevancy** vs. nomic query/document cosine → $\rho = \mathbf{0.71}$ (STRONG)
- **Faithfulness** vs. sentence-level NLI → $\rho = \mathbf{-0.18}$ (judge median 1.000, NLI median 0.000 — they flatly disagree)
- **ContextPrecision** vs. graded NDCG → $\rho \approx \mathbf{0}$ (all variants near-constant on this corpus)

Three takeaways: 1. The judge cost 8h26m; the shortcuts ran in minutes. Two of four metrics can be gated cheaply in CI and only periodically calibrated against a judge. 2. A near-zero correlation is a diagnostic, not a bug — it tells you exactly which judge metric to distrust (here: faithfulness needs a human oracle audit) and which is corpus-degenerate (here: context precision). 3. Run this shortcut-vs-judge check on any new corpus *before* adopting a RAG metric. It's minutes of compute and saves hours of blind trust.

All local: Chroma + nomic-embed-text + gemma4:12b (Ollama) + PyTorch cross-encoders + DeBERTa-v3 NLI. No conversation content in the artifacts — only aggregates, figures, and code.

#RAG #LLM #Evaluation #RAGAS #VectorDatabases #LocalAI